

EXPERIENCE

Embedded SW Engineer, Co-op Semtech (Sierra Wireless) Spring – Winter 2025

- Used Linux, Jira, C++, JavaScript, BitBake, Yocto, and Confluence across Agile and Waterfall workflows.
- Secured an \$XXM deal by delivering critical cellular band mask sw. changes in 3 days to meet customer needs.
- Developed new cellular related software features for 5G enterprise cellular routers and gateways.
- Worked on new OS/UI features using back-end and front-end code to improve CX and fix customer issues.
- Identified and documented new issues on Jira, reviewed and approved team code on GitHub.
- Built OS-level software features for various cellular radio modules used across different routers.

Software Engineer, Intern Abomis Innovations Summer 2024

- Used MeshLab, Visual Studio, Git, and CMake throughout the position to analyze point clouds and program.
- Developed an automated baggage scanning program increasing baggage processing speed by ~45% and detection accuracy by ~20% for airport security using C++, OpenCV, and Open3D.
- Contributed to beta testing and extensive use of the Teledyne Bumblebee X 5GigE camera.

EDUCATION

Vancouver, BC University of British Columbia Fall 2023 – Spring 2027

- BAsC. in Computer Engineering GPA: 3.75 Y2, 3.95 Y3
- Relevant Coursework: C++, Java, Digital Logic Design, Data Structures, Electric Circuits and Power, Embedded Systems Programming, Digital Signal Processing, Software Programming, Computer Hardware.

Languages	Java (OOP)	C++	Python
Hardware Skills	KiCad PCB Design	Arduino Design	3D Image Capturing
Software & Libraries	OpenCV, Open3D	MATLAB	Git, CMake
Computer Knowledge	Computer Architecture	Comm. Protocols	Linux

TECHNICAL EXPERIENCE

Projects

- **Mock Food Delivery App** (2026). Using a dataset, went through entire software design cycle, with agile sprints, and created entire front/backend, complete with API for a food delivery app.
- **FPGA Research @ UBC** (2025). Implemented and simulated sigma-delta modulators on FPGA in Verilog with NCO input and FFT noise analysis.
- **Spotify Statistics Website** (2025). Creating a website that displays user's listening times, top artists, top songs, for different time periods. JavaScript, ReactJS, Spotify API, Figma
- **Wildfire Detection System** (2024). Designed a multi-functional device with multiple different sensors for wildfire monitoring, achieving ~28% improvement in early fire detection. Arduino C/C++
- **PCB Design & Assembly** (2024). Learned PCB design and assembly through an IEEE workshop, developed PCB and circuit designs using KiCad.

ADDITIONAL EXPERIENCE AND AWARDS

- **UBC Dean's Honor Roll Recipient:** Maintained academic average of over 80% while taking 6 courses a term
- **Ninth Prize, UBC SolidWorks Design Project:** 9th prize for Autonomous Car Dealership design from ~80 teams
- **Eagle Scout Award:** Top Boy Scouts of America Award

Languages and Technologies

- C++, Java, JavaScript, Python
- VS Code, Eclipse, ReactJS, Embedded Linux, FastAPI
- OpenCV, Git, CMake, Yocto
- Jira, Confluence, GitHub + Actions